

**BARI PALESE, Italy**WMO: **162700**Lat: **41.138N**Lon: **16.765E**Elev: **54**StdP: **100.68**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.9        | 2.1 | -6.2                          | 2.3 | 5.3  | -4.2 | 2.7 | 5.7  | 11.3                  | 9.1  | 10.0 | 7.9  | 3.7                   | 220  | 0.433 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.3                    | 34.2            | 22.2 | 32.3 | 21.9 | 31.0 | 21.6 | 25.1                | 29.8 | 24.1 | 28.9 | 23.3 | 28.5 | 4.1                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.8                            | 18.8 | 27.7 | 22.8 | 17.6 | 27.2 | 21.6 | 16.4 | 26.7 | 76.7          | 29.6 | 72.8 | 28.8 | 69.4 | 28.6 | 30.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 9.2               | 8.1  | 7.1 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -1.5                       | 38.6 | 1.2                | 2.6 | -2.4                                               | 40.4 | -3.1       | 41.9 | -3.8       | 43.3 | -4.6       | 45.2 |  |  |
|                   |      |     | -2.6                       | 26.5 | 1.3                | 1.2 | -3.6                                               | 27.3 | -4.3       | 28.1 | -5.0       | 28.7 | -5.9       | 29.6 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual       | Jan          | Feb          | Mar          | Apr          | May          | Jun          | Jul          | Aug          | Sep          | Oct          | Nov          | Dec         |
|-------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
|                                                                               |             | (d)          | (e)          | (f)          | (g)          | (h)          | (i)          | (j)          | (k)          | (l)          | (m)          | (n)          | (o)          | (p)         |
| (6) <b>Temperatures, Degree-Days and Degree-Hours</b>                         | DBAvg       | <b>16.3</b>  | <b>8.5</b>   | <b>8.7</b>   | <b>11.0</b>  | <b>14.0</b>  | <b>18.3</b>  | <b>22.8</b>  | <b>25.3</b>  | <b>25.1</b>  | <b>21.1</b>  | <b>17.2</b>  | <b>13.3</b>  | <b>9.8</b>  |
|                                                                               | DBStd       | <b>6.62</b>  | <b>2.78</b>  | <b>3.00</b>  | <b>2.88</b>  | <b>2.76</b>  | <b>2.70</b>  | <b>3.00</b>  | <b>2.48</b>  | <b>2.36</b>  | <b>2.83</b>  | <b>2.72</b>  | <b>3.14</b>  | <b>2.92</b> |
|                                                                               | HDD10.0     | <b>188</b>   | <b>62</b>    | <b>56</b>    | <b>21</b>    | <b>3</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>8</b>     | <b>39</b>   |
|                                                                               | HDD18.3     | <b>1451</b>  | <b>304</b>   | <b>269</b>   | <b>227</b>   | <b>133</b>   | <b>34</b>    | <b>3</b>     | <b>0</b>     | <b>0</b>     | <b>7</b>     | <b>54</b>    | <b>154</b>   | <b>266</b>  |
|                                                                               | CDD10.0     | <b>2488</b>  | <b>17</b>    | <b>19</b>    | <b>52</b>    | <b>123</b>   | <b>258</b>   | <b>383</b>   | <b>474</b>   | <b>469</b>   | <b>334</b>   | <b>223</b>   | <b>106</b>   | <b>31</b>   |
|                                                                               | CDD18.3     | <b>710</b>   | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>3</b>     | <b>34</b>    | <b>135</b>   | <b>216</b>   | <b>211</b>   | <b>91</b>    | <b>19</b>    | <b>2</b>     | <b>0</b>    |
|                                                                               | CDH23.3     | <b>6526</b>  | <b>0</b>     | <b>0</b>     | <b>3</b>     | <b>20</b>    | <b>229</b>   | <b>1219</b>  | <b>2287</b>  | <b>2110</b>  | <b>587</b>   | <b>68</b>    | <b>4</b>     | <b>0</b>    |
|                                                                               | CDH26.7     | <b>2294</b>  | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>4</b>     | <b>42</b>    | <b>404</b>   | <b>904</b>   | <b>789</b>   | <b>146</b>   | <b>6</b>     | <b>0</b>     | <b>0</b>    |
| (14) <b>Wind</b>                                                              | WSAvg       | <b>3.4</b>   | <b>3.5</b>   | <b>3.6</b>   | <b>3.6</b>   | <b>3.4</b>   | <b>3.3</b>   | <b>3.3</b>   | <b>3.5</b>   | <b>3.3</b>   | <b>3.3</b>   | <b>3.1</b>   | <b>3.2</b>   | <b>3.4</b>  |
| (15) <b>Precipitation</b>                                                     | PrecAvg     | <b>548</b>   | <b>49</b>    | <b>54</b>    | <b>49</b>    | <b>40</b>    | <b>35</b>    | <b>26</b>    | <b>26</b>    | <b>30</b>    | <b>58</b>    | <b>60</b>    | <b>54</b>    | <b>62</b>   |
|                                                                               | PrecMax     | <b>848</b>   | <b>117</b>   | <b>136</b>   | <b>128</b>   | <b>131</b>   | <b>111</b>   | <b>97</b>    | <b>98</b>    | <b>127</b>   | <b>217</b>   | <b>168</b>   | <b>144</b>   | <b>125</b>  |
|                                                                               | PrecMin     | <b>319</b>   | <b>3</b>     | <b>6</b>     | <b>6</b>     | <b>2</b>     | <b>2</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>2</b>     | <b>1</b>     | <b>8</b>     | <b>5</b>    |
|                                                                               | PrecStd     | <b>139</b>   | <b>29</b>    | <b>37</b>    | <b>35</b>    | <b>33</b>    | <b>25</b>    | <b>22</b>    | <b>27</b>    | <b>29</b>    | <b>49</b>    | <b>44</b>    | <b>33</b>    | <b>32</b>   |
| (19) <b>Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures</b> | 0.4%        | DB           | <b>18.1</b>  | <b>19.2</b>  | <b>22.2</b>  | <b>25.9</b>  | <b>30.9</b>  | <b>35.0</b>  | <b>37.1</b>  | <b>36.9</b>  | <b>33.0</b>  | <b>27.2</b>  | <b>23.6</b>  | <b>19.2</b> |
|                                                                               |             | MCWB         | <b>12.7</b>  | <b>12.9</b>  | <b>14.4</b>  | <b>16.5</b>  | <b>19.7</b>  | <b>22.3</b>  | <b>22.1</b>  | <b>22.9</b>  | <b>21.4</b>  | <b>19.6</b>  | <b>17.4</b>  | <b>14.4</b> |
|                                                                               | 2%          | DB           | <b>16.1</b>  | <b>17.0</b>  | <b>19.8</b>  | <b>22.8</b>  | <b>27.8</b>  | <b>32.2</b>  | <b>34.5</b>  | <b>33.9</b>  | <b>30.0</b>  | <b>25.2</b>  | <b>21.3</b>  | <b>17.2</b> |
|                                                                               |             | MCWB         | <b>12.3</b>  | <b>12.2</b>  | <b>13.4</b>  | <b>15.5</b>  | <b>18.6</b>  | <b>21.2</b>  | <b>22.0</b>  | <b>22.5</b>  | <b>21.1</b>  | <b>19.1</b>  | <b>16.4</b>  | <b>13.4</b> |
|                                                                               | 5%          | DB           | <b>14.9</b>  | <b>15.3</b>  | <b>17.9</b>  | <b>20.8</b>  | <b>25.9</b>  | <b>30.7</b>  | <b>32.5</b>  | <b>32.0</b>  | <b>28.1</b>  | <b>23.6</b>  | <b>20.0</b>  | <b>16.0</b> |
|                                                                               |             | MCWB         | <b>11.5</b>  | <b>11.3</b>  | <b>12.3</b>  | <b>14.6</b>  | <b>18.0</b>  | <b>20.9</b>  | <b>21.8</b>  | <b>22.6</b>  | <b>20.8</b>  | <b>18.3</b>  | <b>16.0</b>  | <b>12.7</b> |
|                                                                               | 10%         | DB           | <b>13.2</b>  | <b>14.0</b>  | <b>16.2</b>  | <b>19.1</b>  | <b>24.1</b>  | <b>28.9</b>  | <b>31.0</b>  | <b>30.5</b>  | <b>26.5</b>  | <b>22.1</b>  | <b>18.3</b>  | <b>14.6</b> |
|                                                                               |             | MCWB         | <b>10.2</b>  | <b>10.4</b>  | <b>11.8</b>  | <b>13.7</b>  | <b>17.4</b>  | <b>20.8</b>  | <b>21.7</b>  | <b>22.2</b>  | <b>20.3</b>  | <b>17.8</b>  | <b>15.0</b>  | <b>11.6</b> |
| (27) <b>Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures</b> | 0.4%        | WB           | <b>14.1</b>  | <b>14.5</b>  | <b>15.5</b>  | <b>17.7</b>  | <b>21.2</b>  | <b>24.7</b>  | <b>26.2</b>  | <b>27.0</b>  | <b>24.2</b>  | <b>21.7</b>  | <b>18.9</b>  | <b>15.6</b> |
|                                                                               |             | MCDB         | <b>16.9</b>  | <b>17.4</b>  | <b>20.6</b>  | <b>23.0</b>  | <b>25.6</b>  | <b>29.0</b>  | <b>30.1</b>  | <b>31.3</b>  | <b>28.0</b>  | <b>24.9</b>  | <b>21.5</b>  | <b>18.2</b> |
|                                                                               | 2%          | WB           | <b>13.0</b>  | <b>12.9</b>  | <b>14.1</b>  | <b>16.5</b>  | <b>20.0</b>  | <b>23.5</b>  | <b>24.9</b>  | <b>25.5</b>  | <b>22.8</b>  | <b>20.3</b>  | <b>17.5</b>  | <b>14.2</b> |
|                                                                               |             | MCDB         | <b>15.4</b>  | <b>15.9</b>  | <b>17.9</b>  | <b>20.8</b>  | <b>25.4</b>  | <b>28.6</b>  | <b>29.7</b>  | <b>30.3</b>  | <b>27.0</b>  | <b>23.3</b>  | <b>20.3</b>  | <b>16.7</b> |
|                                                                               | 5%          | WB           | <b>11.9</b>  | <b>11.8</b>  | <b>13.2</b>  | <b>15.5</b>  | <b>19.0</b>  | <b>22.6</b>  | <b>23.9</b>  | <b>24.4</b>  | <b>21.9</b>  | <b>19.4</b>  | <b>16.5</b>  | <b>12.9</b> |
|                                                                               |             | MCDB         | <b>14.4</b>  | <b>14.8</b>  | <b>16.6</b>  | <b>19.3</b>  | <b>24.5</b>  | <b>28.1</b>  | <b>29.1</b>  | <b>29.2</b>  | <b>26.2</b>  | <b>22.5</b>  | <b>19.1</b>  | <b>15.3</b> |
|                                                                               | 10%         | WB           | <b>10.6</b>  | <b>10.7</b>  | <b>12.4</b>  | <b>14.5</b>  | <b>18.1</b>  | <b>21.6</b>  | <b>23.0</b>  | <b>23.5</b>  | <b>21.0</b>  | <b>18.5</b>  | <b>15.5</b>  | <b>11.9</b> |
|                                                                               |             | MCDB         | <b>13.0</b>  | <b>13.5</b>  | <b>15.7</b>  | <b>18.1</b>  | <b>23.1</b>  | <b>27.2</b>  | <b>28.9</b>  | <b>28.5</b>  | <b>25.4</b>  | <b>21.6</b>  | <b>18.1</b>  | <b>14.2</b> |
| (35) <b>Mean Daily Temperature Range</b>                                      | 5% DB       | MDBR         | <b>7.3</b>   | <b>7.8</b>   | <b>8.4</b>   | <b>8.8</b>   | <b>9.4</b>   | <b>9.4</b>   | <b>9.3</b>   | <b>9.2</b>   | <b>8.6</b>   | <b>8.2</b>   | <b>7.7</b>   | <b>7.3</b>  |
|                                                                               |             | MCDBR        | <b>9.1</b>   | <b>9.7</b>   | <b>11.1</b>  | <b>11.4</b>  | <b>12.1</b>  | <b>12.3</b>  | <b>12.1</b>  | <b>11.8</b>  | <b>10.8</b>  | <b>10.0</b>  | <b>9.0</b>   | <b>8.7</b>  |
|                                                                               | 5% WB       | MCWBR        | <b>6.0</b>   | <b>6.1</b>   | <b>6.6</b>   | <b>6.4</b>   | <b>6.3</b>   | <b>6.4</b>   | <b>7.0</b>   | <b>7.0</b>   | <b>5.9</b>   | <b>5.7</b>   | <b>5.6</b>   | <b>5.7</b>  |
|                                                                               |             | MCDBR        | <b>8.0</b>   | <b>8.7</b>   | <b>9.9</b>   | <b>9.9</b>   | <b>10.6</b>  | <b>10.2</b>  | <b>9.8</b>   | <b>9.7</b>   | <b>9.5</b>   | <b>9.0</b>   | <b>7.7</b>   | <b>7.8</b>  |
|                                                                               |             | MCWBR        | <b>5.7</b>   | <b>5.7</b>   | <b>6.4</b>   | <b>6.1</b>   | <b>6.2</b>   | <b>6.5</b>   | <b>7.0</b>   | <b>7.0</b>   | <b>5.9</b>   | <b>5.8</b>   | <b>5.3</b>   | <b>5.6</b>  |
| (40) <b>Clear-Sky Solar Irradiance</b>                                        | taub        | <b>0.340</b> | <b>0.363</b> | <b>0.398</b> | <b>0.445</b> | <b>0.441</b> | <b>0.431</b> | <b>0.426</b> | <b>0.432</b> | <b>0.423</b> | <b>0.394</b> | <b>0.366</b> | <b>0.342</b> |             |
|                                                                               | taud        | <b>2.457</b> | <b>2.371</b> | <b>2.274</b> | <b>2.147</b> | <b>2.224</b> | <b>2.293</b> | <b>2.315</b> | <b>2.300</b> | <b>2.319</b> | <b>2.383</b> | <b>2.414</b> | <b>2.449</b> |             |
|                                                                               | Ebn at Noon | <b>809</b>   | <b>840</b>   | <b>846</b>   | <b>828</b>   | <b>841</b>   | <b>847</b>   | <b>848</b>   | <b>832</b>   | <b>812</b>   | <b>794</b>   | <b>766</b>   | <b>770</b>   |             |
|                                                                               | Edh at Noon | <b>79</b>    | <b>98</b>    | <b>121</b>   | <b>147</b>   | <b>139</b>   | <b>130</b>   | <b>126</b>   | <b>125</b>   | <b>115</b>   | <b>96</b>    | <b>82</b>    | <b>74</b>    |             |
| (44) <b>All-Sky Solar Radiation</b>                                           | RadAvg      | <b>1.80</b>  | <b>2.55</b>  | <b>3.87</b>  | <b>5.10</b>  | <b>6.24</b>  | <b>7.02</b>  | <b>7.14</b>  | <b>6.26</b>  | <b>4.58</b>  | <b>3.20</b>  | <b>2.05</b>  | <b>1.60</b>  |             |
| (45) <b>RadStd</b>                                                            |             | <b>0.19</b>  | <b>0.23</b>  | <b>0.39</b>  | <b>0.39</b>  | <b>0.38</b>  | <b>0.35</b>  | <b>0.31</b>  | <b>0.36</b>  | <b>0.27</b>  | <b>0.25</b>  | <b>0.16</b>  | <b>0.20</b>  |             |

**Historical Trends**

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|                         |  | +0.49 | N/A     | N/A    | N/A     | -0.86 | -1.35 | N/A         | -102    | +160    | N/A     |
| Regional (11 neighbors) |  | +0.33 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -67     | +92     | N/A     |